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IT-Enabled Coordination for Ambidextrous Interorganizational Relationships

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Contextual ambidexterity of an interorganizational relationship (IOR) is the ability of its management system to align partners' activities and resources for short-term goals and adapt partners' cognitions and actions for long-term viability. It is an alternative to structural ambidexterity in which separate units of the IOR pursue short- and long-term goals. We theorize that when utilized to coordinate the IOR, information technology (IT)-enabled operations and sensemaking, along with interdependent decision making, promote the IOR's contextual ambidexterity. We test our hypotheses on both sides of a customer-vendor relationship using data collected from (1) the account executives of one of the world's largest supply chain vendors ($n = 76$) and (2) its customers ($n = 238$). We find commonalities and differences in the influence coordination mechanisms have on contextual ambidexterity from the vendor's and the customer's perspectives. For both customers and vendors, contextual ambidexterity improves the quality and performance of the relationship, and decision interdependence promotes contextual ambidexterity. For customers, using operations support systems (OSSs) and interpretation support systems (ISSs) enhances contextual ambidexterity. For vendors, the impact of both OSS use and ISS use on contextual ambidexterity depends on the duration of the relationship. Our study shows that IT-enabled operations and sensemaking are key enablers of IOR ambidexterity and that vendors should combine these IT capabilities with relationship-specific knowledge that accumulates with relationship duration.

Keywords: interorganizational relationships; interorganizational systems; contextual ambidexterity; coordination; decision interdependence; sensemaking; operations support systems; interpretation support systems; relationship duration

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1. Introduction

In hypercompetitive environments, firms are not only challenged to align decisions, resources, and activities with partners, but also to coevolve strategically with changes in partners' requirements and the environment (Tanriverdi et al. 2010). However, firms' inability to coevolve their interorganizational relationships (IORs) with partners' requirements and environmental turbulence challenges their ability to create value for partners (Lewin et al. 1999), leading more than 50% of alliances to fail (Gulati et al. 2012). For an IOR to remain viable in the long run, it needs to be ambidextrous by synergistically pursuing (1) *alignment*, or coherence in partners' activities and resources to accomplish relationship objectives; and (2) *adaptation*, or evolution of partners' cognitions, actions, and activities with changes in the environment and partners' objectives (Lavie and Rosenkopf 2006, Im and Rai 2008).

With the need for IORs to be ambidextrous, partnering firms are challenged to establish an organizational design to pursue alignment and adaptation synergistically. Given the complexity of implementing disparate structures for alignment and adaptation, along with the difficulty of reintegrating the activities and outcomes of these disparate units, contextual ambidexterity is an alternative, promising organizational design (Im and Rai 2008). It is the IOR-management system's culture, processes, and routines that encourage the synergistic pursuit of paradoxical activities by partners without requiring disparate structural arrangements (Gibson and Birkinshaw 2004). We focus our study on identifying the information technology (IT)-enabled coordination mechanisms, along with the decision-making structures, that can enable contextual ambidexterity in IORs (Malone 1987, Gulati and Singh 1998).

Past work on interorganizational systems (IOS) has shown that IT-enabled interfirm *operations* (e.g., integrated supply chain planning and execution; visibility system deployment; collaborative planning, forecasting, and replenishment) promote both the coherence and adaptation of interfirm operational processes (Kumar and van Dissel 1996, Rai et al. 2006, Simchi-Levi et al. 2008, Rai et al. 2012). With advances in business intelligence systems, IT can also be used to support partners' *sensemaking* of the relationship and its environment (Malhotra et al. 2005). IT can be leveraged to scan the environment, identify trends and opportunities, and support distributed problem solving. As such, IT capabilities for sensemaking can be used to develop shared mental models and promote mutual understanding between partners, plausibly influencing the IOR's ability to achieve both alignment and adaptation. Because IT capabilities can support both interfirm *operations* and partners' *sensemaking* about the relationship and its environment, we conceptualize the use of two types of information systems (IS)—namely, operations support systems (OSSs) and interpretation support systems (ISSs)—as aspects of an IOR's coordination. Although OSS use integrates activities and decisions across partners, ISS use promotes partners' discovery and learning about the IOR and its environment (Boland et al. 1994, Malhotra et al. 2005, Robey et al. 2008). In addition, as interdependence in partners' decision making facilitates knowledge and information exchange between partners and enables them to evaluate decision choices pertaining to the relationship more holistically (Chatfield and Yetton 2000, Gulati and Sytch 2007, Rai et al. 2012), we consider its influence on contextual ambidexterity. Accordingly, we examine the influence of OSS use, ISS use, and decision interdependence on contextual ambidexterity, leading to our first research question: How does an IOR's use of IT-enabled coordination for operations (i.e., OSS use) and sensemaking (i.e., ISS use) and its use of interdependent decision making influence the IOR's contextual ambidexterity?

An IOR develops assets and rigidities with time. On the asset side, it develops rich communication and collaboration routines (Uzzi 1997), leading to the accumulation of partner-specific knowledge (Hult et al. 2004). On the rigidities side, it risks becoming inertial, constraining its ability to adjust to changes in partners' requirements and the environment (Sorensen and Stuart 2000). As the IOR's effectiveness in utilizing IT-enabled coordination and interdependent decision making is likely to be contingent on partner-specific knowledge that accumulate over time, we examine how IOR duration moderates the influence of the use of OSSs, ISSs, and decision interdependence on contextual ambidexterity, leading

to our second research question: How does an IOR's relationship duration moderate the influence of its use of IT-enabled coordination for operations (i.e., OSS use) and sensemaking (i.e., ISS use) and its use of interdependent decision making on the IOR's contextual ambidexterity?

As contextual ambidexterity is an alternative to other ambidexterity mechanisms, it is important to understand how it affects key outcomes in different contexts (Raisch and Birkinshaw 2008). There is some evidence that has accumulated in this regard. In a study of business units from 10 multinational firms across nine industries, Gibson and Birkinshaw (2004) found contextual ambidexterity to enhance business units' performance. Also, in a study of IORs for supply chain services, Im and Rai (2008) found contextual ambidexterity to enhance exploitative and exploratory knowledge sharing, thus promoting sustained IOR performance benefits for both partners. Elaborating on Im and Rai (2008), we examine how contextual ambidexterity affects both the performance and quality of the relationship, leading to our third research question: What is the impact of contextual ambidexterity on relationship performance and relationship quality?

In developing our theoretical logic and model, we contrast contextual ambidexterity from structural ambidexterity, highlighting the comparative advantages of contextual ambidexterity. We draw on coordination theory (Van de Ven et al. 1976, Malone 1987) to argue why IT-enabled interfirm coordination at both the operational and sensemaking levels and interfirm decision interdependence will influence contextual ambidexterity, why these effects will change as the IOR ages, and why contextual ambidexterity will affect the IOR's performance and quality. We conducted our empirical study in the context of the U.S. logistics industry and collected dyadic data on customer-vendor relationships from one of the largest global supply chain vendors and its customers. Our results contribute to our understanding of the relative emphasis customers and vendors should place on IT-enabled coordination mechanisms for operations and sensemaking and on joint decision making to increase the relationship's performance and quality.

2. Theoretical Background

2.1. Contextual Ambidexterity of IORs

There are two key types of ambidexterity mechanisms—structural and contextual. With respect to structural ambidexterity, two views have been advanced. The first view suggests creating separate business units to manage paradoxical activities simultaneously within an organization (Tushman and O'Reilly 1996). Both

units are spatially separated and internally inconsistent and have their own culture, processes, and routines for stability or innovation. They are loosely coupled internally, but they eventually integrate at a higher management level. The second view suggests transforming units that maintain stability into units that innovate, and vice versa, over time as necessary (Duncan 1976, Siggelkow and Levinthal 2003). This temporal switching leads to paradoxical activities being accomplished over time.

The structural view of ambidexterity requires organizations to incur costs not only for establishing and maintaining disparate structures but also for harmonizing these disparate structures' strategies and outcomes over time (Lewis 2000). As such, establishing structural arrangements requires the creation of separate units for contradictory activities and developing reintegration mechanisms to unify these units' contradictory activities (O'Reilly and Tushman 2004). In the IOR context, the structural perspective not only requires a plural system of governance within each partnering firm but also requires the structural integration of the governance units for alignment and adaptation across the partnering firms (Heide 2003). Because integrating governance units for alignment and adaptation across firm boundaries will likely require each partnering firm to make changes in authority, procedures, and routines, partners are unlikely to be motivated to adopt structural integration for governance (Puranam et al. 2009).

As an alternative to the structural view of ambidexterity, the contextual view focuses on the management system—namely, the culture, processes, and routines—to achieve synergy between alignment and adaptation. Gibson and Birkinshaw (2004) defined contextual ambidexterity as a management system's capacity to pursue both alignment and adaptation synergistically. They noted that a management system could not only enable the pursuit of both alignment and adaptation but could also enable their synergistic integration. Drawing on Gibson and Birkinshaw's (2004) conceptualization, we define an IOR's contextual ambidexterity as the capacity of its management system to achieve synergies between the alignment and adaptation of goals, resources, and activities for the interfirm relationship.

With its focus on achieving synergies between aligning partners' goals, resources, and activities and modifying them with changes in partners' conditions and the environment, contextual ambidexterity should enable IORs to proactively and reactively embrace change (Conboy 2009). By establishing an integrated system focusing on the IOR's current mode of creating value and unfolding changes in partners' conditions and the environment, the management system emphasizes synergies between what is

being accomplished and what *can* be accomplished. For example, Hewlett-Packard and United Parcel Service (UPS) have collaborated not only to effectively achieve short-term objectives but also to evaluate new opportunities for value cocreation by leveraging past experiences and accumulated resources and capabilities (Lewis and Rai 2006, Lewis et al. 2007, Rai et al. 2012).

2.2. IOR Coordination Structure as Enabling Contextual Ambidexterity

We focus on the characteristics of an IOR's coordination structure as enablers of its contextual ambidexterity. An IOR's coordination structure integrates interfirm resources and activities to accomplish relationship goals (Van de Ven et al. 1976). As coordination mechanisms establish the process for the IOR's decision making and information processing (Malone 1987), they are not only instrumental in executing day-to-day activities but also in influencing short-term improvements and long-term opportunities. We extend Gibson and Birkinshaw's contextual ambidexterity framework by focusing on the effects three aspects of an IOR's coordination structure have on its contextual ambidexterity, as described below.

First, we examine the influence of IT-enabled operational coordination—specifically the utilization of OSSs—on contextual ambidexterity. OSS use refers to using IT to structure and integrate interfirm operational routines in a relationship. It requires mapping interdependent tasks between partners into identifiable chunks of activities (Kumar and van Dissel 1996), enabling partners to better understand each other's information needs to efficiently execute interfirm processes (Malhotra et al. 2005). By structuring interfirm operations, OSS use reduces process- and technological-level barriers for information and knowledge sharing between partners (Nahapiet and Ghoshal 1998). It provides visibility to existing routines and errors, thereby enabling partners to discover problems and adjust routines and technology for short-term improvements and long-term opportunities (Rai et al. 2012). Given these capabilities, OSS use can enable partners to integrate and adjust their operations (Malone 1987, Rai et al. 2006, Bala and Venkatesh 2007) and thus influence their ability to achieve the relationship's current and future operational goals.

Second, we examine the impact of IT-enabled sensemaking—specifically, ISS use—on contextual ambidexterity. ISS use pertains to scanning, processing, and analyzing existing and new information and to refining partners' cognitive frameworks on the relationship's goals, routines, and resources. To facilitate partners' interpretation of situational elements, ISSs support processing massive amounts of data

and multiple representations of information, mining data, and generating diverse views (Malhotra et al. 2005, Rai et al. 2009a). It also facilitates the interpretation of events, uncertainties, and irregularities surrounding the IOR, thereby enhancing partners' mutual understanding about the trends, patterns, and types of changes they face. These sensemaking capabilities should enable partners to better understand how the present goals, activities, and resources can facilitate or constrain adaptation and create synergies between alignment and adaptation. FedEx's Quick Response Solutions is a good example of the impacts IT-based coordination has on IOR ambidexterity. FedEx developed Quick Response Solutions to address the challenges of delivering time-sensitive components to customers when equipment failure occurs (Jasper 2006). FedEx established three stocking locations (i.e., the customer site, forward stocking facilities, and a global distribution center) to minimize customers' inventory carrying costs. It deployed OSSs to provide visibility of stored components and coordinate operational activities across locations, and it also provided decision aids to technicians on the best sourcing and routing method to replace failed components. With the help of this system, technicians were able to perform daily operations without disruption and flexibly deal with changes in supply availability at various locations, thus enhancing customer satisfaction. Leading logistics vendors (e.g., UPS, FedEx) have also established systems to trace key service-level metrics and diagnose problems and opportunities for improvement (Lewis and Rai 2006, Rai et al. 2012) that, in turn, promote relationship adaptation.

Finally, we consider the influence of the IOR's decision-making structure on contextual ambidexterity. An IOR's decision-making structure allocates decision rights to partners and integrates decision

activities across partners to accomplish relationship goals (Gulati and Singh 1998). Contextual ambidexterity requires partners to understand the interdependencies in decision choices with respect to relationship outcomes (Levinthal 1997). (See Appendix A for additional details on decision interdependence.) As interdependent decision making enables partners to pool information and expertise and jointly evaluate plausible relationship outcomes (Uzzi 1997), we consider it an essential property of an IOR's coordination structure that should affect its contextual ambidexterity.

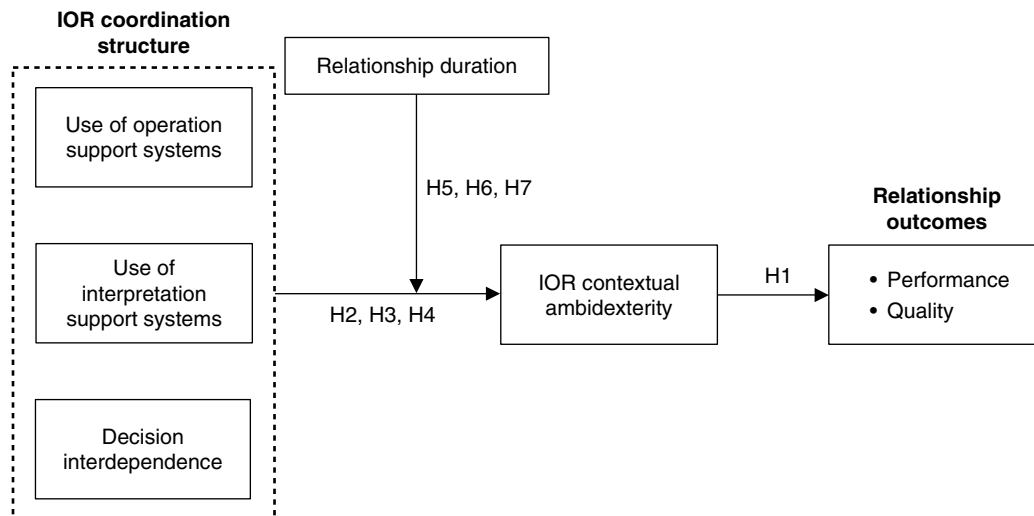
3. Research Model

We hypothesize the influence of (1) an IOR's contextual ambidexterity on relationship outcomes, (2) an IOR's coordination structure on its contextual ambidexterity, and (3) relationship duration in moderating the effects of an IOR's coordination structure on its contextual ambidexterity (Figure 1).

3.1. Relationship Outcomes

We suggest two reasons that contextual ambidexterity will improve relationship performance (Raisch and Birkinshaw 2008). First, with "either/or" strategies emphasizing one aspect of contextual ambidexterity, partners are likely to overlook the other aspect. Alignment supports short-term activities, whereas adaptation promotes long-term initiatives. To sustain performance, partners cannot only focus on alignment as the current mode of creating value may not be effective in new situations. Second, strict adherence to a cookbook approach or the free pursuit of new arrangements does not provide the means to pursue both short-term challenges and long-term opportunities. The synergistic pursuit of alignment and adaptation enables partners to combine short-term

Figure 1 Predictors and Outcomes of IOR Contextual Ambidexterity



objectives with emergent goals, market opportunities, and changes in the external environment. By establishing a management system that synergistically reinforces alignment and adaptation, IORs should generate performance benefits from both exploiting partners' competencies and exploring value cocreation opportunities.

Further, we suggest that contextual ambidexterity will enhance relationship quality for two reasons. First, partners are less likely to face ambiguity and conflict in dealing with both present problems and future challenges when the IOR management system focuses on synergies between them (Gibson and Birkinshaw 2004). Second, partners are more likely to feel that the interfirm relationship is fulfilling when it is not focused either on returns that are near term and certain or on returns that are far term and futuristic, but when it combines both these pursuits (Johnson et al. 2004).

In sum, contextual ambidexterity should establish an IOR management system that orients the partnering firms to cocreate value effectively, both in the short- and long-term, thereby leading to superior relationship outcomes (Grover and Kohli 2012, Rai et al. 2012). This leads us to hypothesize the following:

HYPOTHESIS 1 (H1). *An IOR's contextual ambidexterity promotes its relationship performance and quality.*

3.2. Operations Support System Use

An IOR's use of OSSs refers to its reliance on ISs for routine interfirm operations that coordinate partners' actions. These interfirm routines are repetitive patterns of actions among partners that connect partners and help them to develop a shared understanding about their interdependent tasks (Feldman and Rafaeli 2002). Traditionally, such routines have been considered sources of stability, a conceptualization emphasizing the structural aspect of routines. However, an alternative conceptualization depicts routines as generative systems that support both stability and change depending on the circumstances the actors face (Feldman and Pentland 2003, Turner and Rindova 2012). Therefore, an agency (e.g., people at a time in a place associated with an IOR) can engage in change and innovation and deal with challenges and opportunities by using effective routines that have been assembled.

Supporting the stability aspect of routines, OSSs structure partners' actions by utilizing the automation capabilities of IOSs (e.g., electronic data interchange, automated shipping, and error correction). These systems are instrumental in aligning partners actions by defining complex event-action dependencies, clarifying partners' roles and responsibilities, and automating the execution of interfirm processes (e.g., information exchange between partners). Second,

they enable partners to initiate changes in interfirm routines (Pentland and Feldman 2007). They provide an impetus to change routines by enabling the granular tracking of activities and surfacing problems when the desired result is not generated, an unexpected result is obtained, or the result unveils new possibilities (Feldman 2000). Moreover, open-standard IOSs (e.g., electronic business eXtensible Markup Language and RosettaNet) enable partnering firms—dominant and nondominant firms alike—to overcome rigidities in interfirm routines and to change interfirm process flexibly (Bala and Venkatesh 2007). In fact, open-standard IOSs, characterized by their openness, expandability, and interoperability, help refine existing routines and lower reconfiguration costs (Zhu et al. 2006, Robey et al. 2008). Using these inherent flexibilities in OSSs, partners can adapt to impetuses for change by flexibly responding to the circumstances, reprioritizing tasks, and reconfiguring existing resources and routines. In sum, OSS use should facilitate contextual ambidexterity, leading us to hypothesize the following:

HYPOTHESIS 2 (H2). *OSS use in an IOR has a positive effect on its contextual ambidexterity.*

3.3. Interpretation Support System Use

The use of ISSs refers to the IOR's reliance on ISs to support the interpretation and manipulation of information for managerial processes and decision making (Malhotra et al. 2005). ISS features include communicating, storing, indexing, searching, and matching (Goodman and Darr 1998, Chou 2003); data processing and analytics applied to massive volumes of transaction data (Vandenbosch and Higgins 1996); and generating multiple interpretations. Typically, ISSs combine data warehousing with analytics to support the generation of multiple interpretations (Malhotra et al. 2005).

As IORs are open social systems that need to seek, process, and use information from external sources for survival, they should not only be able to obtain information from the environment and each partner, but they should also be capable of refining information through the interpretation processes that translate events, develop models, extract meaning, and assemble concepts (Daft and Weick 1984). According to Boland et al. (1994), ISs used to facilitate interpretation can be designed with six principles (i.e., ownership, easy travel, multiplicity, indeterminacy, emergence, and mixed forms) to accomplish their interpretation tasks by enabling rich forms of representation and dialogue among actors with varying perspectives.

We posit that ISS use facilitates the synergistic pursuit of alignment and adaptation between partners. Our rationale is that ISSs enable partners to generate

multiple frames for complex situations, thus enabling them “to make rich representations of their understanding, reflect upon those representations, engage in dialogue about them with others, and use them to inform action” (Boland et al. 1994, p. 457). The creation of diverse representations for understanding opens the possibility of developing relational cognitive maps (i.e., managerial perceptions) (Huber 1990, Thomas et al. 1993) that accommodate contradictory elements, open partners up to new opportunities, and form new interactions among partners (Boland et al. 2007).

Furthermore, ISS use should help the partnering firms in an IOR frame problems to strengthen or challenge existing mental maps as well as solve problems over the course of the interfirm collaboration (Vandenbosch and Higgins 1996). When ISS use generates emergent framing that fits existing cognitive maps reasonably well, the existing cognitive elements can be adjusted to achieve better alignment with the framing (Todorova and Durisin 2007). However, when emergent framing conflicts with existing cognitive maps, it may be necessary for partners to create new cognitive maps to accommodate the new framing. In the first case, the emergent framing is assimilated to improve existing cognitive maps, whereas in the latter case, the new framing is used to transform existing cognitive maps (Piaget 1952). Although improving cognitive maps will guide partners’ refinement of ongoing activities to achieve current IOR goals, establishing new cognitive maps will help partners’ explore new ways to cocreate value given changing partner requirements and the environment (Malhotra et al. 2007). Because ISS use can create synergies in the refinement and development of cognitive maps, it should support contextual ambidexterity in IORs, leading us to hypothesize the following:

HYPOTHESIS 3 (H3). *ISS use in an IOR has a positive effect on its contextual ambidexterity.*

3.4. Decision Interdependence

Decision interdependence refers to the degree of interaction on decision choices between partnering firms in an IOR (Siggelkow and Levinthal 2003). Past work has examined how decision choices interact to influence organizational outcomes (Simon 1962, Thompson 1967, Duncan 1972) and joint action among interfirm partners (Gulati and Sytch 2007). A key implication from past studies is that when partners’ decision choices map to a rugged performance landscape, they should establish interdependent decision making to discover dependencies in decision choices and avert local optima traps (Levinthal 1997).

In our case, contextual ambidexterity, by definition, requires IORs to establish a management system that considers how partners’ decision choices facilitate

or hurt synergies between current alignment actions and future adaptation moves. As such, contextual ambidexterity is a management solution for an IOR performance landscape that is multi peaked with multiple local optima (i.e., high performance variability) (Levinthal 1997). Elaborating, an IOR’s current actions are likely to influence future options because of path dependence (e.g., the switching costs of partners can be influenced by the choice of interfirm process standards) (Levinthal 1997). Moreover, one partner’s decisions are likely to influence the other’s decisions such that overlooking interdependencies results in a rugged performance landscape not being navigated effectively (e.g., revision in a manufacturer’s plan to launch a product has implications for how its supply chain partner should execute logistics and overlooking these implications degrades the IOR’s performance) (Klein et al. 2007).

As such, interdependent decision making will enable partners in the relationship to pursue distant searches (i.e., global evaluations of alternatives), go beyond local optima, and evaluate the broad solution space to enhance the chances of long-term relationship survival (Levinthal 1997, Selnes and Sallis 2003). This does not mean that the relationship can forgo local searches for its long-term survival; on the contrary, the relationship still requires local searches to take advantage of knowledge and skills embedded in established routines and partner policies (Levinthal 1997, Gavetti and Levinthal 2000). In sum, partners’ decision interdependence is likely to reduce structural and attitudinal gaps (Gulati and Sytch 2007) on how short-term decisions relate to long-term possibilities, thus enabling the IOR to achieve synergies between alignment and adaptation. Therefore, we hypothesize the following:

HYPOTHESIS 4 (H4). *Decision interdependence between partners in an IOR has a positive effect on its contextual ambidexterity.*

3.5. Moderating Roles of Relationship Duration

We suggest that IORs accumulate relational assets over time and that these assets complement the effects of the three coordination properties on contextual ambidexterity. First, we suggest that the positive impact of OSS use on contextual ambidexterity will be greater when relationship duration is high. The stock of routines developed through OSS use between partners should accumulate over time, providing partners with a richer arsenal to deal with external stimuli (Levinthal and Rerup 2006). That is, a richer stock of routines will help partners react to external stimuli and prepare for actions more quickly. With a richer set of routines, partners can also recombine established routines better to create new routines. For example, a supplier and retailer that have

been in a long-term relationship and have independent forecasting systems are likely to find it easier to combine their existing systems to build a single forecasting system for collaborative planning in comparison to a supplier and retailer who have just initiated a relationship (Arshinder et al. 2008). Relationship duration should also help partners accumulate experience and competencies to establish and revise systems and processes for tackling external stimuli (Levinthal and Rerup 2006). For example, once partners learn how to utilize service-oriented architectures and Web services to support routines for addressing customer needs, they will have an easier time revising routines for emergent customer needs. This enrichment in the stock of routines and in experience and competencies to establish and revise systems and processes should enable IORs of longer durations to leverage OSS use more effectively to achieve synergies between alignment and adaptation, leading us to hypothesize the following:

HYPOTHESIS 5 (H5). *The positive effect of OSS use in an IOR on its contextual ambidexterity is moderated by relationship duration such that the effect is stronger for a relationship of longer duration.*

Second, we suggest that relationship duration should amplify the positive effect of ISS use on the IOR's contextual ambidexterity. Longer relationship duration should expand relational memory through repeated interactions and partner-specific experiences (Hult et al. 2004). When partners possess relational memory, they realize that it will support better decision making for the relationship (Hult et al. 2004). Also, relational memory and knowledge acquisition form a deviation-amplifying loop wherein one promotes more of the other, expanding the IOR's pool of information and knowledge with time. As a result, partners in long-term relationships will complement the ISS's analytical capabilities with the expanded base of relational memory and knowledge, thereby promoting the pursuit of synergies between alignment and adaptation. Thus, we propose the following hypothesis:

HYPOTHESIS 6 (H6). *The positive effect of ISS use in an IOR on its contextual ambidexterity is moderated by relationship duration such that the effect is stronger for a relationship of longer duration.*

Finally, we suggest that relationship duration should amplify the positive influence of decision interdependence on contextual ambidexterity. As relationship duration helps partners develop "routines of negotiation and mutual adjustment" (Uzzi 1997, p. 47) and institutionalize sophisticated problem-solving mechanisms (Larson 1992), it should enable

partners to address problems in a timely and flexible manner (Uzzi 1997, Gulati and Sych 2007). Consequently, IORs of longer durations should be positioned well to leverage decision interdependence to negotiate the tension between alignment and adaptation and to achieve synergies between them. With longer relationship duration, partners are likely to commit to joint goals (Gulati and Sych 2007), enabling them to leverage decision interdependence for achieving synergies between alignment and adaptation. These characteristics of longer relationship duration should result in decision interdependence having a greater positive influence on contextual ambidexterity for relationships of longer durations, leading us to hypothesize the following:

HYPOTHESIS 7 (H7). *The positive effect of decision interdependence between partners in an IOR on its contextual ambidexterity is moderated by relationship duration such that the effect is stronger for a relationship of longer duration.*

4. Research Design

4.1. Sample and Data Collection

We chose the logistics industry in the United States as the sampling frame because it aligns well with our focus on IORs, ambidexterity, and IT's role in IOR coordination for multiple reasons. First, the logistics industry accounts for significant economic activity with estimates suggesting that the industry's revenues for North America are \$119 billion (Coyle et al. 2009). As third-party logistics (3PL) vendors (e.g., UPS, FedEx, DHL-Exel) have expanded their services to include procurement, fulfillment, warehouse management, and customer support, IORs with 3PL vendors have emerged to be crucial for the success of firms' supply chains (Simchi-Levi et al. 2008). As such, customer firms in an IOR with a 3PL vendor may experience declines in market share, cost overruns, and relationship satisfaction if the interdependencies in interfirm logistics are not aligned and adapted to customers' strategic plans and objectives (Rai and Sambamurthy 2006, Rai et al. 2012). Second, the logistics industry has been expanding to cope with increasing customer needs for globalization, agility, and operational efficiency (Coyle et al. 2009). Logistics vendors confront high uncertainty from changing customer requirements for supply chain services and also from shifting competitive dynamics. Finally, logistics vendors have been investing aggressively in developing IT-enabled capabilities for logistics operations and business intelligence (Langley et al. 2007), making IT salient in IOR coordination. These characteristics emphasize the importance for IORs in the logistics industry to achieve synergies

between alignment and adaptation, and IT's influential role in this pursuit.

Our unit of analysis was the IOR between a 3PL vendor's business unit and a customer's business unit for supply chain services. We used a matched-pairs sampling strategy to understand the phenomenon from the perspective of both the vendor and customer. This research design enabled us to compare the impact of the coordination mechanisms on contextual ambidexterity from the vendor and customer perspectives and to uncover commonalities and differences in this regard. Our sample consisted of IORs between one of the world's largest supply chain vendors and its customers. We selected two divisions at the focal vendor firm: one that provides only freight services and the other that provides an integrated suite of supply chain solutions, including freight services, warehousing, and inventory management. The services offered by these two divisions provide variation in the complexity of supply chain services and a basis to evaluate the stability of results after controlling for variance in the complexity of the supply chain services exchanged in a vendor-customer relationship.

We followed the recommended procedures to collect reliable and valid data on dyadic relationships and used key informants on both sides of the relationship (John and Reve 1982). Given our focus on the management of customer-vendor relationships, we determined the account managers at the customer firm and at the vendor firm to be the most informed respondents because of their direct involvement in managing the relationship. Data were first collected from account managers at the customer firm after which the corresponding account managers at the focal vendor firm were solicited to provide matching data.

Two waves of solicitation emails were sent to the account managers of 3,379 target customer relationships. A total of 331 customers participated in the online survey for a response rate of 10%; however, only 238 customer surveys were usable for the analysis. Among these, 171 customers received support from a dedicated vendor account manager, and the rest were supported by a shared call center operated by the vendor. After the customer data were collected, the vendor version of the survey was sent to the vendor account managers who were responsible for the 171 customer relationships. In all, 81 vendor account managers provided the matching data for a response rate of 47%. Five of these responses had significant missing data, resulting in 76 usable vendor account manager surveys for our analysis. Nonresponse bias was assessed by comparing data from early and late survey respondents across the customer and vendor samples (Armstrong and Overton 1977). As no significant differences were found for any of the independent variables across the early and late respondents,

we concluded that nonresponse bias was not a serious concern in our data.

Customer informants consisted of senior executives (19% of the 238), mid-level managers (67%), and analysts (14%). On average, the customer informants had worked for their companies for 18 years, and the vendor informants had worked for their companies for 14.4 years. These informants were involved in IOR management for an average of 7.7 years (customer) and 4.6 years (vendor). Therefore, we concluded that the informants had sufficient knowledge to supply reliable data for this study. The average duration of the customer-vendor relationship was 7.3 years, which was greater than the average relationship duration of 2.98 years across all types of 3PL contracts in the logistics industry (Langley et al. 2007). Moreover, the vendor was a substantial partner to its customers in that it was responsible for an average of 38% of the customer's total contract volume for the supply chain service involved.

4.2. Measures

We developed the vendor and customer versions of the questionnaires following recommended procedures (Straub 1989, MacKenzie et al. 2011). We took several steps to ensure that our measures were reliable and valid and that they were meaningfully specified for our empirical study's context. We developed a deep contextual understanding of the empirical setting through extensive interactions with the vendor and its customers. One of the authors has worked closely with the vendor over a period of 10 years on a research program directed at understanding the management of supply chain relationships. This research program has involved interactions with and surveys of executives at the vendor firm who are responsible for customer relationship management, IT support, business development, marketing, and sales. In addition, it has involved interactions with executives at customer firms across major industries (e.g., automotive, high-tech, retail, and healthcare) responsible for sourcing logistics services and managing relationships with logistics vendors.

Based on the contextual background of the vendor and its customers and an extensive review of the literatures on organizational ambidexterity, IOS, and IORs, we developed preliminary versions of the questionnaires. Where feasible, we adapted existing scale items to our investigative context. Preliminary versions of the questionnaires were refined via four pilot tests using a policy-capturing method (Webster and Trevino 1995) and in-depth interviews with account managers at the vendor firm. Measures for relationship outcomes (i.e., relationship performance and relational quality) and for contextual and IOR coordination (i.e., OSS use, ISS use, and decision interdependence) were measured on a seven-point Likert

scale (1 = strongly disagree and 7 = strongly agree). We now elaborate on our measures for the constructs, which are shown in Appendix B.

Relationship performance. We refined prior relationship performance measures (Selnes and Sallis 2003, Subramani 2004) to represent partners' performance benefits from the logistics IOR. We identified the following performance elements that encompassed efficiency and effectiveness and applied to both customers and vendors: (1) operating costs, (2) revenue, (3) service quality, (4) order fulfillment time, and (5) new product introduction speed. We measured the extent to which partners realized these performance benefits from the IOR in the past two years and computed a mean composite index of the realized benefits.

Relationship quality. Relationship quality has different aspects (e.g., trust, success, power), and its emphasis varies depending on the relationship context (Naude and Buttle 2000). In general, relationships that generate successful outcomes are considered to be valuable and typically last longer. Accordingly, we used an omnibus measure of relationship quality based on the mean value of two items evaluating perceived relationship success and satisfaction over the past two years (Johnson et al. 2004).

Contextual ambidexterity. Ambidexterity has been measured by multiplying scales for components posited to be synergistic. For example, Gibson and Birkinshaw (2004) measured management systems' contextual ambidexterity by multiplying the measures for management systems' alignment and adaptability. Similarly, He and Wong (2004) and Cao et al. (2009) measured learning ambidexterity by multiplying the measures for exploitation and exploration, which were posited to be two synergistic forms of learning. The espoused rationale for this multiplicative approach has been that it recognizes that the underlying components are not necessarily competitive over scarce resources but are complementary and mutually reinforcing. This rationale implies that, with an ambidextrous management system, an IOR's alignment between partners (e.g., joint pursuit of efficiency, coherence, and consistency) reinforces its adaptation to new products and markets (e.g., flexibility, agility, and reconfiguration), and vice versa. We measured IORs' alignment and adaptability by refining Gibson and Birkinshaw's (2004) measures for alignment and adaptability to our context and developing three-item scales for each. Leading into these survey questions, we provided illustrations of the management system elements (e.g., service-level agreements, information exchange for planning and review, incentives, contract renegotiation) for logistics IORs that were meaningful to respondents. We measured contextual ambidexterity as the mean of nine

cross-product items—i.e., the cross products of the three alignment (CAL) items with the three adaptability (CAD) items. As such, our multiplicative measure for contextual ambidexterity (ranging from 1 to 49 as both the alignment and adaptability scales range from 1 to 7) is consistent with the notion that IORs' alignment and adaptation are mutually reinforcing.

Decision interdependence. Drawing on the literature on decision making and interfirm relationship learning (Duncan 1972, Selnes and Sallis 2003), we created a three-item scale to measure decision interdependence between IOR partners. These items captured the degree of decision interdependence with respect to interaction, information sharing, and collaboration for decision making in the relationship. We used a mean value of these items to measure decision interdependence between partners in an IOR.

Use of operations support systems. We used the mean value of three items to capture IS use to support an IOR's operations. Our specification of the items was informed by the literatures on coordinating IORs in general (e.g., Selnes and Sallis 2003, Tippins and Sohi 2003) and IT-enabled IOR coordination (e.g., Rai et al. 2006, Rai and Tang 2010, Venkatesh and Bala 2012), which emphasize the role of IT in establishing systems to execute operational activities, coordinate related activities, and achieve compliance with process standards.

Use of interpretation support systems. Drawing on Malhotra et al. (2005), we developed a four-item scale capturing ISS use for inferences from past events, new/multiple information interpretations, and multiple information representations. These items represent the ISS characteristics for processing large quantities of data and helping partners uncover hidden patterns and trends. We used the mean value of these items as a measure of ISS use in an IOR.

Relationship duration. We measured relationship duration in years since the relationship was initiated, which is consistent with past studies on IORs (e.g., Im and Rai 2008, Rai et al. 2012).

Control variables. We included customer firm size, environmental uncertainty, relationship goals, customer transaction volume, and supply chain solution complexity as control variables for contextual ambidexterity as well as relationship performance and relationship quality. It was important to control for *customer firm size* because larger customer firms are likely to have richer resources to foster alignment and adaptability activities with their partners and to influence relationship conditions for greater performance. Customer firm size was measured using the total number of employees in the customer firm (Swamidass and Kotha 1998). *Environmental uncertainty* was included as a control variable as firms in the relationship can increase their development

of contextually ambidexterity to address volatility (Jansen et al. 2005). We carefully designed the wording of these items to reflect the changing environment for customer preferences, technologies, and markets that affect both partners in the relationships. Informants were also asked to evaluate the uncertainty of the environment in which the relationship operated. We controlled for *relationship goal for value cocreation* as firms can orient goals to cocreate value rather than limit them to transactional exchange (Jap 1999, Rai et al. 2012). As partnering firms can cocreate value by improving existing solutions or discovering innovative solutions, we used two items to assess customer/vendor perspectives on the extent to which the relationship's goals are oriented to cocreate value using each of the two approaches. We used the mean value of these two items to measure relationship goal for value cocreation. We also controlled for *customer transaction volume*, which was the vendor's percentage share of the customer's wallet for outsourced logistics services (Du et al. 2007) as a greater share of the customer's outsourced logistics services may lead to increased support by the vendor (Sheth and Shah 2003). This increased support can improve the collaborative climate in the relationship, thereby promoting the development of contextual ambidexterity and the attainment of positive relationship outcomes. Finally, we controlled for *supply chain solution complexity* by using a dummy variable capturing whether the relationship was associated with the vendor's division that provided only freight services (0 = low complexity) or an integrated suite of supply chain services (1 = high complexity) (Otto and Kotzab 2003).

5. Results

5.1. Descriptive Statistics, Measurement Validation and Common Method Bias Assessment

Our data contained 8.34 (3.46)% missing values in the customer (vendor) sample. Based on nonsignificant p -values for the Little's missing completely at random (MCAR) test for both samples, we concluded that the data were missing completely at random. Accordingly, we employed the multiple imputation missing value replacement technique according to suggested procedures (Rubin 1987, Schafer and Olsen 1998). Table 1 presents the descriptive statistics and correlations among the constructs for the vendor and customer samples.

The scale reliabilities, average variance extracted, and item-to-construct correlations for multi-item reflective measures (i.e., all constructs with the exception of relationship performance that was measured as a composite of five aspects of performance) suggested good measurement properties. (See Appendix B.) For each of these constructs, Cronbach's alpha was

greater than 0.72, composite reliability was greater than 0.83, and average variance extracted was greater than 0.62, exceeding the recommended thresholds for these criteria (Fornell and Larcker 1981, Hair et al. 2010). All indicators revealed strong principal component loading on their hypothesized factors and did not show high cross-loadings on other factors. (See Appendix C.) To test for discriminant validity, we employed a more stringent test of shared variance between two constructs (Fornell and Larcker 1981). The results for both the customer and vendor samples showed that the squared correlations between the constructs did not exceed the average variance extracted for each construct, indicating discriminant validity. The results collectively provide evidence that the measures were reliable and valid.

We conducted two tests to examine if common method bias was a concern in our samples. First, as suggested by Podsakoff et al. (2003), we evaluated the degree of common method bias by including a method factor in the measurement model and associating all indicators in the model with this factor. We found that, on average, the substantive constructs accounted for 0.907 (0.845) of the variance and that the method factor, on average, explained only 0.005 (0.020) of the variance in the customer (vendor) sample, suggesting that common method bias was not a serious problem with our data. Second, we employed the marker variable approach (Lindell and Whitney 2001). The marker variable is supposed to be theoretically unrelated to the focal variables and show low correlations with any of them. We selected the respondent's managerial level (senior executive, mid-level-manager, account analyst) as the marker variable. Adjusting for the marker variable, all correlations among the substantive variables revealed no change in magnitude (rounding up to two decimal places) for the customer sample and dropped only by 0.02 for the vendor sample, and all significant correlations remained significant at the same level. These results suggest that common method bias is not a concern with our data.

5.2. Hypothesis Testing

We employed ordinary least squares (OLS) regression to test the hypotheses. To mitigate multicollinearity when testing for interaction effects, we mean centered the predictor variables (i.e., OSS use, ISS use, decision interdependence) and the moderator variable (i.e., relationship duration) (Aiken and West 1991). Tables 2 and 3 summarize the results of the regression analyses for the customer and vendor samples.¹ Contextual

¹ We replicated the analysis with the subsample of 171 customers who received support from a dedicated vendor account manager as opposed to a shared call center and observed the same pattern of results as reported with the full sample of 238 customers.

Table 1 Descriptive Statistics and Correlations

Variable	1	2	3	4	5	6	7	8	9	10	11	12
1. Relationship Performance	—	0.69	0.47	0.37	0.12	0.13	−0.08	−0.03	0.01	0.03	0.14	0.05
2. Relationship Quality	0.70	—	0.48	0.50	0.20	0.38	−0.01	0.15	−0.15	0.07	−0.03	0.24
3. Contextual Ambidexterity	0.69	0.65	—	0.39	0.28	0.58	−0.08	0.64	−0.08	−0.04	−0.35	0.11
4. Decision Interdependence	0.36	0.36	0.47	—	0.29	0.13	−0.05	0.26	0.02	−0.09	0.13	0.19
5. OSS Use	0.38	0.35	0.46	0.44	—	0.52	0.01	0.45	−0.19	0.00	−0.05	0.15
6. ISS Use	0.45	0.34	0.47	0.42	0.61	—	0.08	0.56	−0.14	0.10	−0.23	0.19
7. Firm Size	0.08	−0.03	0.10	0.19	0.18	0.16	—	−0.12	−0.09	−0.05	0.22	0.02
8. Environmental Uncertainty	0.17	0.21	0.22	0.34	0.29	0.36	0.08	—	−0.10	−0.05	−0.24	0.26
9. Relationship Goal	0.33	0.34	0.42	0.58	0.35	0.33	0.06	0.41	—	−0.03	0.14	−0.03
10. Customer Transaction Volume	0.06	0.14	0.17	0.22	0.17	0.04	−0.07	−0.04	0.18	—	−0.03	−0.03
11. Relationship Duration	0.00	−0.01	−0.01	0.09	0.08	0.06	0.16	0.05	0.12	0.07	—	0.23
12. SC Solutions Complexity	0.03	0.06	−0.01	−0.16	−0.06	−0.01	−0.04	0.02	−0.20	−0.16	0.05	—
Mean	4.18 (4.79)	5.15 (5.82)	21.77 (28.6)	4.87 (5.02)	4.93 (5.72)	4.21 (4.76)	1,788 (2,112)	4.35 (5.35)	5.45 (5.95)	38.07 (37.5)	7.28 (6.87)	—
S.D.	1.30 (0.86)	1.54 (0.94)	11.55 (9.2)	1.23 (0.36)	1.37 (0.10)	1.27 (1.03)	3,358 (3,772)	1.14 (1.06)	1.23 (0.46)	34.00 (35.0)	4.12 (3.67)	—

Notes. 1. Customer sample ($N = 238$) information below the diagonal; vendor sample ($N = 76$) information above the diagonal. 2. All correlations greater than or equal to 0.16 are significant at the 0.05 level (two-tailed). 3. Descriptive statistics are reported for both the customer sample and the vendor sample (in parentheses).

ambidexterity explains significant variance in relationship performance (adjusted $R^2 = 0.50$ [0.35], $p < 0.01$ for customers [vendor]) and in relationship quality (adjusted $R^2 = 0.44$ [0.30], $p < 0.01$) for customers [vendor]) and has a significant positive effect on relationship performance as well as relationship quality, thus supporting H1 (Table 2). The predictors explain significant variance in contextual ambidexterity (adjusted $R^2 = 0.35$ [0.55], $p < 0.01$) (Table 3). OSS use has a mixed effect on contextual ambidexterity (positive in the customer sample and negative in the vendor sample), thereby partially supporting H2. Consistent with H3, ISS use has a positive effect on contextual ambidexterity in both samples. Supporting H4, decision interdependence has a positive effect on contextual ambidexterity in both the

customer and vendor samples. The interaction effect of relationship duration and OSS use on contextual ambidexterity was significant in the vendor sample but not in the customer sample, thus partially supporting H5. Specifically, in the vendor sample, OSS use *decreases* contextual ambidexterity when relationship duration is low (i.e., $p < 0.05$ when relationship duration is -1 SD below the mean) and *increases* contextual ambidexterity when relationship duration is very high ($p < 0.05$ when relationship duration is $+2$ SD above the mean; $p > 0.05$ when relationship duration is $+1$ SD above the mean). (See Figure 2.) Similarly, the interaction effect of relationship duration and ISS use on contextual ambidexterity was significant in the vendor sample but not in the customer sample, thereby partially supporting H6. Specifically,

Table 2 Results of the OLS Regression Analysis for Relationship Outcomes

DV	Customer sample		Vendor sample	
	Relationship performance	Relationship quality	Relationship performance	Relationship quality
<i>Controls</i>				
(Constant)	4.02** (0.19)	4.79** (0.24)	4.76** (0.26)	5.20** (0.30)
Firm size	0.01 (0.03)	−0.06* (0.03)	−0.06* (0.04)	−0.01 (0.04)
Environmental uncertainty	−0.02 (0.06)	0.07 (0.07)	−0.36** (0.09)	−0.27** (0.10)
Relationship goal	0.07 (0.06)	0.10+ (0.08)	−0.07 (0.18)	−0.29+ (0.20)
Customer transaction volume	−0.01+ (0.00)	0.01 (0.00)	0.01 (0.01)	0.01 (0.01)
Relationship duration	0.01 (0.01)	−0.01 (0.02)	0.08** (0.03)	0.03 (0.03)
SC solutions complexity	0.11 (0.12)	0.24+ (0.16)	0.02 (0.18)	0.43* (0.20)
<i>Direct effects</i>				
H1: Contextual ambidexterity	0.08** (0.01)	0.08** (0.01)	0.08** (0.01)	0.07** (0.01)
Adjusted R^2	0.501	0.435	0.348	0.298

Notes. 1. Unstandardized coefficients are shown with standard errors in parentheses. 2. + $p < 0.05$, * $p < 0.05$, ** $p < 0.01$ for one-tailed tests given directional hypotheses.

Table 3 Results of the OLS Regression Analysis for Contextual Ambidexterity

DV = Contextual ambidexterity	Customer sample			Vendor sample		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
<i>Controls</i>						
(Constant)	18.43** (2.19)	18.61** (1.97)	18.57** (2.00)	28.30** (2.70)	29.81** (2.36)	30.64** (2.20)
Firm size	0.50 ⁺ (0.30)	0.05 (0.28)	0.08 (0.28)	0.16 (0.36)	−0.01 (0.32)	−0.02 (0.30)
Environmental uncertainty	0.58 (0.66)	−0.65 (0.62)	−0.63 (0.62)	4.52** (0.75)	2.92** (0.80)	2.71** (0.74)
Relationship goal	3.71** (0.63)	1.89** (0.65)	1.82** (0.66)	0.35 (1.85)	0.01 (1.62)	−0.40 (1.55)
Cust. transaction volume	0.04* (0.02)	0.02 (0.02)	0.03 ⁺ (0.02)	−0.02 (0.12)	−0.05 (0.10)	−0.03 (0.10)
Relationship duration	−0.26 ⁺ (0.17)	−0.26* (0.15)	−0.27* (0.15)	−0.56** (0.25)	−0.53** (0.22)	−0.02 (0.26)
SC solutions complexity	2.24 ⁺ (1.40)	2.12* (1.26)	2.15* (1.28)	0.21 (1.80)	−0.84 (1.57)	−1.08 (1.46)
<i>Direct effects</i>						
H2: OSS use		1.47** (0.59)	1.40** (0.59)		−1.70* (0.90)	−1.06 (0.86)
H3: ISS use		2.16** (0.63)	2.23** (0.64)		3.40** (0.94)	3.50** (0.88)
H4: Decision interdependence		1.89** (0.66)	1.96** (0.67)		5.05** (1.32)	6.10** (1.33)
<i>Interaction effects</i>						
H5: OSS use × RD			−0.11 (0.14)			0.44* (0.22)
H6: ISS use × RD			−0.06 (0.13)			0.36* (0.20)
H7: DI × RD			0.15 (0.15)			−0.35 (0.27)
Adj. R^2	0.189	0.345	0.341	0.401	0.551	0.615
ΔR^2		0.156**	—		0.150**	0.064**

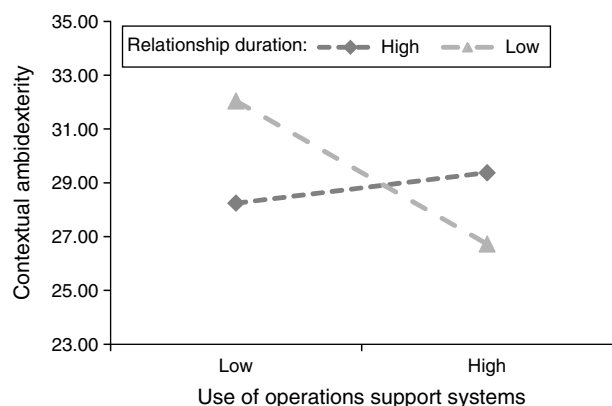
Notes. 1. Unstandardized coefficients are shown with standard errors in parentheses. 2. ⁺ $p < 0.05$, * $p < 0.05$, ** $p < 0.01$ for one-tailed tests given directional hypotheses. 3. RD: Relationship Duration; DI: Decision Interdependence.

from the vendor's perspective, we found relationship duration to amplify the positive effect of ISS use on contextual ambidexterity ($p < 0.05$ for the significance of the simple slopes when relationship duration is -1 SD or $+1$ SD from the mean). (See Figure 3.) Contrary to H7, relationship duration was not observed to moderate the effect of decision interdependence on contextual ambidexterity in either the customer or vendor sample.

Finally, we evaluated if contextual ambidexterity mediated the following effects on relationship outcomes: (1) the direct effects of the three coordination variables (i.e., decision interdependence, OSS use, and ISS use) and (2) the interaction effects of

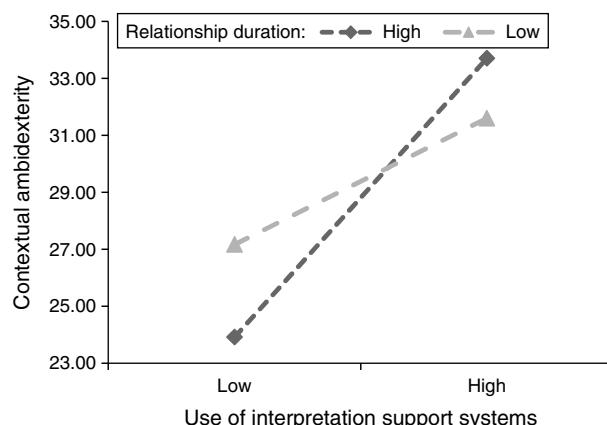
the IT-enabled coordination variables (i.e., ISS use and OSS use) with relationship duration (Baron and Kenny 1986, Muller et al. 2005). We used the Sobel's test to evaluate these mediation effects (MacKinnon et al. 2002) and found all the examined mediation effects to be significant at $p < 0.05$. Specifically, for the customer sample, we found contextual ambidexterity fully mediated the effects of the coordination variables on relationship outcomes. (The interaction effects were not significant in the customer sample, so we did not evaluate the mediation of the interactions.) For the vendor sample, we found contextual ambidexterity (1) fully mediated the direct effects of the

Figure 2 Moderation Effect of Relationship Duration for Operations Support System Use^a



^aLow and high refer to -1 SD below the mean and $+1$ SD above the mean, respectively.

Figure 3 Moderation Effect of Relationship Duration for Interpretation Support System Use^a



^aLow and high refer to -1 SD below the mean and $+1$ SD above the mean, respectively.

three coordination variables on relationship performance and relationship quality and (2) fully mediated the interaction effects of the IT-enabled coordination variables (i.e., OSSs and ISSs) with relationship duration on relational performance and relationship quality. The results collectively provide strong evidence that coordination mechanisms lead to superior relationship outcomes by enhancing IORs' contextual ambidexterity.

6. Discussion

6.1. Summary of Findings

We found contextual ambidexterity to enhance relationship performance and relationship quality from both the customer and vendor perspectives. We also found relationship duration to have a negative effect on contextual ambidexterity and the three coordination mechanisms to influence contextual ambidexterity—directly or interactively with relationship duration—from both the vendor and customer perspectives. Our findings indicate that OSS use fosters contextual ambidexterity on the customer side, but they reveal that this effect is negative on the vendor side and only becomes positive when relationship duration is high. In contrast, our results indicate that ISS use promotes contextual ambidexterity from the perspective of both partners. We found the positive effect of ISS use on contextual ambidexterity becomes stronger for the vendor but not for the customer as the relationship ages. Our results also show that interdependent decision making enhances IORs' contextual ambidexterity from the vendor and customer perspectives, although we did not find relationship duration to moderate the effect of decision interdependence on contextual ambidexterity in either the customer or vendor sample.

Our findings have important theoretical implications for the ambidextrous management of IORs and the role of IT capabilities in this regard. They provide insights into how contextual ambidexterity can enable IORs to coevolve with changes in partners' requirements and the environment, how IT capabilities and decision structures for IOR coordination can promote ambidexterity, and how IT capabilities for IOR coordination can be used to mitigate the risk of inertia as IORs age. We elaborate on the theoretical implications and follow with a discussion on the practical implications, limitations, and future research.

6.2. Theoretical Implications

6.2.1. Developing Contextual Ambidexterity for Relationship Outcomes. Our study contributes to the growing literature on IT-enabled IORs and the performance implications of IOR management systems and IT capabilities. Our findings that contextual

ambidexterity enhances relationship performance and relationship quality for both the customer and vendor provide evidence that IORs are more successful when they are capable of coevolving with the environment (Rai and Tang 2010, Tanriverdi et al. 2010). These results suggest that by establishing an IOR management system to pursue alignment and adaptation synergistically, contextual ambidexterity represents a powerful organizational design that can serve as an alternative to structural separation, where conflicting objectives are relegated to segregated organizational units coordinated through an overlay of reintegration mechanisms. We suggest that contextual ambidexterity leads both the customer and vendor in an IOR to enjoy greater performance benefits and higher relationship quality because it enables the IOR management system to (1) align partners' resources and actions in pursuit of relationship objectives and (2) adapt the relationship to changes in partners' objectives, factor markets, and the broader environment.

Our study adds to the understanding of the contexts in which ambidexterity is beneficial (Inkpen and Tsang 2008, Raisch and Birkinshaw 2008) and how partnering firms can cocreate value (Grover and Kohli 2012, Rai et al. 2012). Our findings suggest that an ambidextrous organizational design is valuable not only in managing a firm's internal resources (e.g., individuals and business units) but also in leveraging external resources, specifically IORs for core products and services. Given that the vendor in our study was a leader in the concentrated logistics industry, the results show that IOR ambidexterity accrues performance benefits and increases relationship quality for both partnering firms even when the vendor is an industry leader with vast internal resources. The overall implication is that as IORs become a crucial part of a firm's strategy, the ambidextrous management of IORs will need to be developed as a capability to cocreate value with partners.

Our results also provide insights into why aging IORs face challenges in remaining beneficial for both partners (Im and Rai 2008, Lewis et al. 2007). The finding that IORs' contextual ambidexterity declines with relationship duration suggests that IORs' capacity to synergistically pursue current objectives and explore future opportunities decreases over time. Thus, whereas past work has revealed that partner-specific assets accrue as IORs age (e.g., shared understanding, cooperative norms) (Hult et al. 2004), our study shows that IT-enabled coordination capabilities for operations support and interpretation support gain importance over time in overcoming the suppressive effect of relationship age on contextual ambidexterity.

6.2.2. Enabling Contextual Ambidexterity with IT Capabilities for IOR Coordination. Our study contributes to the IOS literature by identifying the IT capabilities that enable partnering firms in an IOR to achieve synergies between alignment and adaptation. It expands the IOS literature's predominant focus on IT's role in achieving alignment (e.g., Truman 2000, Mukhopadhyay and Kekre 2002) or adaptation (e.g., Gosain et al. 2004) to understanding IT's role in pursuing alignment and adaptation synergistically with partners. Past studies have shown how IOS (e.g., electronic data interchange and supply chain automation) can enable partners to align activities and enhance transaction efficiency (e.g., shorter order-completion cycle, fewer order-processing errors), and a few studies have also examined how IOS can promote flexibility, adaptation, and sensemaking (Gosain et al. 2004, Wang et al. 2006). Extending these studies, our findings show that IS use for both operations and sensemaking influences contextual ambidexterity in IORs, with interesting differences in these effects between the customer and vendor.

Role of IT-Enabled Operations. Our findings extend the literature on IOS that has evaluated how operational systems can affect relationship outcomes (e.g., Mukhopadhyay and Kekre 2002, Malhotra et al. 2005, Klein and Rai 2009, Rai et al. 2012, Bala and Venkatesh 2007). They reveal that the influence of OSS use on relationship outcomes is mediated by contextual ambidexterity, thereby adding to our understanding of the mechanisms through which OSS use creates benefits in IORs. They also show that the effects of OSS use on contextual ambidexterity and, consequently, relationship outcomes are contingent on both the party in the IOR (i.e., the customer or vendor) and the duration of the relationship, as we now elaborate.

From the customer's vantage point, our findings reveal that OSS use enhances contextual ambidexterity regardless of relationship age. These results suggest that customers perceive OSS use as important for IORs to achieve stability and adapt to change. This finding is generally consistent with past studies that have found IT-enabled operations (e.g., order entry, storage/retrieval of operational information, process traces, automated rules to coordinate activities) to not only facilitate the integration of activities/decisions to achieve current operational objectives but also to provide the visibility and flexibility necessary to uncover and pursue opportunities (Tarafdar and Gordon 2007, Doherty and Terry 2009).

In contrast, from the vendor's perspective, our findings reveal that the effect of OSS use on contextual ambidexterity is moderated by relationship duration (Figure 2). We suggest that OSS use that goes beyond transactional exchange requires relationship-specific investments in systems and processes, placing

a greater resource burden and holdup risk on the vendor relative to the customer. Also, the vendor's understanding of the potential complementary resources and capabilities (e.g., knowledge about the customer's systems and processes) that can be leveraged along with OSS use is likely to develop with relationship history through strategic communications with the customer (Rai et al. 2012). As a result, when relationship history is limited, the vendor is neither likely to make idiosyncratic investments in the customer nor is it likely to be able to leverage relationship-specific complementary assets, thus leading OSS use to have a negative influence on contextual ambidexterity for the vendor in the IOR. However, as mutual commitment, trust, and relationship-specific knowledge develop over time, the vendor should be better positioned to integrate its systems and processes with those of the customer and to leverage accumulated relationship-specific assets, thereby realizing OSSs' potential for contextual ambidexterity.

Role of IT-Enabled Sensemaking. Our study suggests that IT-enabled coordination, including ISS use, enables partnering firms to enhance relationship-specific absorptive capacity—namely, organizational routines and processes for acquiring, assimilating, transforming, and exploiting knowledge in an IOR (Malhotra et al. 2005, 2007; Roberts et al. 2012). Our results support the position that IORs using ISSs to a greater extent have a greater ability to achieve synergies between alignment and adaptation, thereby increasing relationship performance and relationship quality for both customers and vendors. We suggest that ISSs enable partners in the IOR to critically analyze historical data, combine information across internal/external sources in the customer and vendor firms, integrate perspectives of multiple stakeholders in the customer and vendor firms, and represent information effectively to reveal patterns and challenge partnering assumptions. As such, ISS use helps partners generate business intelligence to align activities, resources, and decisions for the pursuit of current goals and to adapt activities, resources, and goals based on emergent problems and opportunities.

Moreover, we found relationship duration to moderate the positive effect of ISS use on contextual ambidexterity from the perspective of the vendor (Figure 3). Our finding in the vendor sample that ISS use has a positive effect on contextual ambidexterity when relationship duration is low suggests that ISS use generates insights for the vendor in terms of how to achieve synergies between alignment and adaptation even in the early stages of the relationship. Over time, the contextual knowledge about the relationship that accumulates through repeated encounters (Dyer and Chu 2000, Rai et al. 2012) becomes a valuable resource for the vendor that complements

the positive effects of ISS use on contextual ambidexterity. As such, vendors responsible for providing services are able to achieve synergies between alignment and adaptation when they have had long interaction histories with customers *and* when they use a rich set of interpretation support features to generate business intelligence. Indeed, as relationship duration increases, customers are likely to expect vendors to take greater initiative in creating both short- and long-term value, especially when the market offers alternative supply sources (Pfeffer and Salancik 2003). For instance, our follow-up discussions with some of the vendors' long-term customers indicated that they expect the vendor to be proactive in interpreting the supply chain solutions environment and in generating ideas not only for improving current solutions but also for developing innovative solutions that better address customers' changing business needs for supply chain services.

6.2.3. Role of Decision Interdependence. Our research contributes to our understanding of how decision interdependence between IOR partners can develop relationships, avoid destructive conflicts, and adapt the relationship to partners' changing needs and capabilities (Gulati and Sytch 2007, Palmatier et al. 2007). Specifically, we found that decision interdependence—which fosters joint decision making, information sharing, and the joint evaluation of choices—promotes contextual ambidexterity from the perspectives of both the customer and vendor. Interdependent decision making provides the context for the customer and vendor to exchange information (Rai et al. 2009b), learn about partners' broader goals and objectives, and develop shared schemata about the relationship to interpret partners' current and emergent needs (Malhotra et al. 2005). Consequently, interdependent decision making should expand the IOR management system's ability to not only pursue alignment and adaptation but also to achieve synergies between them, thereby resulting in greater relationship performance and relationship quality. Given that we did not detect relationship duration to moderate the effect of decision interdependence on contextual ambidexterity, the synergies between alignment and adaptation are promoted by establishing interdependent decision making between partners early in the relationship and maintaining this interdependent decision making in the relationship as it matures.

6.3. Limitations and Future Research

Our study has its limitations and provides avenues for future research. First, our empirical study was conducted in the context of the logistics industry, and the relationships in our sample were limited to the customers of one large logistics vendor. Although there were variations in IOR coordination

across the customer-vendor relationships in our sample, future research should evaluate the generalizability of our results across multiple vendors and customers in different industry contexts. To investigate industry effects, studies should take into account the differences in the material resource environment (e.g., market turbulence and technological turbulence) and the institutional environment (e.g., institutional practices and belief systems) across industries (Chiasson and Davidson 2005).

Second, although past work has examined the impacts of partner interdependence on relationship performance (Palmatier et al. 2007), we need to understand how partner interdependence affects the pursuit of contextual ambidexterity in IORs. For example, dependence asymmetry between partners may reduce trust and commitment, thereby undermining efforts to promote ambidexterity and hampering relationship outcomes. In contrast, joint dependence reflects mutual economic actions, generates a high level of interaction between partners, and requires each partner to pay attention to the other's needs (Gulati and Sytch 2007), all of which should foster ambidexterity. Thus, future research needs to elaborate our understanding of how partner interdependence in an IOR influences ambidexterity.

Third, future research can investigate other aspects of an IOR's coordination structure (Stinchcombe 1985). For example, how should partners' design authority systems, create standard operating procedures, and dispute resolution procedures across a relationship? Conceivably, these hierarchical elements contribute to task coordination and information processing in an IOR (Gulati and Singh 1998) and can impact contextual ambidexterity.

Fourth, as our study shows, IOSs have differential impacts on contextual ambidexterity for the vendor (who is a technology leader in the industry) and its customers. Future research can examine these differences further by considering how partners' asymmetric IOS objectives and investments for operations and sensemaking influence the development of IT capabilities to coordinate IORs. Large vendors are likely to prioritize IOS optimization and standardization across their diverse customer base, whereas customers are likely to prioritize IOS integration and flexibility that accommodate their changing market circumstances. Understanding how tensions in partners' IOS objectives and investment priorities affect the evolution of IT capabilities to coordinate an IOR and the mechanisms through which value is cocreated by partners will be important extensions to the emergent literature on IT-enabled value cocreation (e.g., Grover and Kohli 2012, Rai et al. 2012, Sarker et al. 2012).

Finally, our results show that contextual ambidexterity can be enhanced when partners do not limit the

use of ISs to short-term goals or operational support. One crucial antecedent to IS use is partners' interpretations of technology—called technological frames—which affect “people's assumptions, expectations, and knowledge about the purpose, context, importance, and role of technology” (Orlikowski and Gash 1994, p. 179). Technological frames promote technology use when they clarify ambiguities about the technology and provide a guide for using it to solve problems and make decisions (Gioia 1986). They can also constrain technology use when they unduly strengthen established assumptions and limit the creative use of technology for new and unstructured problems. Future research can investigate how technological frames influence partners' IS use across the operations and sensemaking levels in ways that create synergies between alignment and adaptation.

6.4. Practical Implications

Our study provides insights into how managers at customer and vendor firms can achieve synergies between alignment and adaptability (or ambidexterity) and achieve favorable relationship outcomes. First and foremost, managers must recognize that there is a credible risk for IORs to become overly focused on alignment for the pursuit of current goals without regard to how these actions adversely affect future options for partners in an IOR to cocreate value. They need to directly confront this risk by conceiving a management system that achieves synergies between alignment and adaptation on an ongoing basis by evaluating the path dependencies between current actions and future options.

Once managers have crossed the chasm from limiting attention to alignment or treating alignment and adaptation as independent choices to conceiving an ambidextrous management system for an IOR, they need to establish IOR coordination mechanisms that encompass operations, sensemaking, and decision making. They should evaluate the decision-making structures that are being used in IORs and assess the interdependencies among partners' decision choices. Specifically, managers need to establish routines for information sharing and joint decision making as these practices will enable them to better understand and manage interdependencies in decision choices, thus promoting synergies between alignment and adaptation. Importantly, managers should expand the roles of ISs to encompass both operations and sensemaking, and they need to safeguard against the risk that IT's role for sensemaking will be overlooked and limited to operations. Managers can improve the alignment and adaptation of IORs by using ISs to generate inferences from past events, combine and

mine information from disparate sources, and generate and share views from different stakeholders' perspectives. Moreover, vendor managers need to place greater emphasis on OSS use and ISS use as relationship duration increases. Such IS use with long-term customers can promote greater synergies between alignment and adaptability than with short-term customers and can counter the inertia that is likely to accumulate as relationships age.

Relationship managers at customer and vendor firms will need to develop the coordination capabilities used in an IOR. One way to jointly develop these capabilities is establishing interaction procedures that enable key stakeholders from customer and vendor firms to collaborate on key decisions related to the coordination capabilities of the IOR. These capabilities can include face-to-face periodic reviews among stakeholders as well as regular interactions between account managers. Similarly, promoting OSS use and ISS use will require collaboration among personnel in different roles (e.g., IS, business development, operations) at the vendor and customer firms to determine what operational routines should be stable or flexible, what information should be pooled across firms, what analyses can generate insights into how to improve alignment, what views should be generated to discover how alignment constrains adaptation, and how views across stakeholders in the vendor and customer firms should be debated and reconciled.

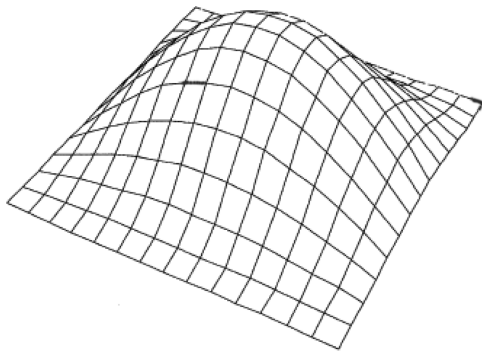
7. Conclusion

Contextual ambidexterity enables IORs to achieve performance benefits and enhance relationship quality from the perspectives of both customers and vendors. IT-enabled operations and sensemaking, along with interdependent decision making, are key coordination mechanisms promoting contextual ambidexterity, with notable differences in customer and vendor perspectives. From the customer's perspective, IT-enabled coordination mechanisms and interdependent decision making promote contextual ambidexterity. From the vendor's perspective, IORs, over time, accumulate relationship-specific resources that are instrumental in leveraging OSSs and ISSs to achieve synergies between alignment and adaptation. Overall, our findings suggest that IS use at both the operations and sensemaking levels plays a major role in enabling contextual ambidexterity and increasing the value derived from IORs for both the vendor and customer.

Acknowledgments

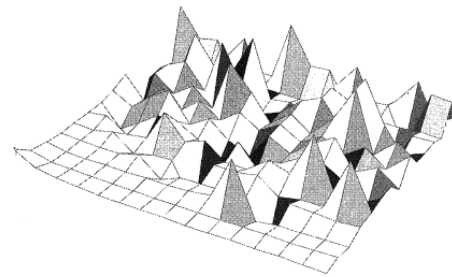
The authors contributed equally, and their names appear in alphabetical order. The authors gratefully acknowledge the support provided by the vendor firm. They also acknowledge the valuable comments received from the senior editor, Viswanath Venkatesh; the associate editor, Anne Massey; and the anonymous reviewers.

Figure A.1 Smooth and Single-Peaked Landscape



Notes. Diagrams of smooth fitness landscape (p. 119) from (Gavetti G, Levinthal D (2000) Looking forward and looking backward: Cognitive and experiential search. *Administrative Science Quarterly* 45(1):113–137).

Figure A.2 Rugged and Multi-peaked Landscape



Notes. Diagrams of rugged landscape (p. 120) from (Gavetti G, Levinthal D (2000) Looking forward and looking backward: Cognitive and experiential search. *Administrative Science Quarterly* 45(1):113–137).

Appendix A. Decision Interdependence

Our theoretical arguments with respect to IOR decision interdependence rely on the NK framework developed in evolutionary biology (Davis et al. 2007). This framework is characterized by a set of N decision choices and K interactions among the decisions. The performance (or fitness) landscape is the trajectory of performance values created by combining every possible decision choices.

Figure A.1 illustrates a smooth landscape in which decision choices (N) do not have many interactions (low K) with

each other and the performance landscape is smooth and highly correlated. In this case, the performance landscape has a single peak and partners can identify a single optimal solution. In contrast, Figure A.2 shows a rugged landscape in which the performance landscape has multiple peaks (local optima), because decision choices have many interactions (high K) and a single change in decision choice significantly influences the performance value of other choices (Gavetti and Levinthal 2000).

Appendix B. Measurement Items for Constructs

Constructs	Measurement items
Relationship performance	By working with this partner, [vendor/our business unit] has received the following benefits over the past two years: 1. Reduced operational cost for [vendor/business unit] operations. 2. Increased revenue. 3. Improved service quality. 4. Faster order fulfillment. 5. Improvement in the launch of new products.
Relationship quality	How would you rate your relationship with [this customer/vendor] in overall terms over the last two years? 1. For the most part, this relationship is very successful for us. 2. Overall, this relationship is very satisfactory for us. (Customer: CA = 0.94, CR = 0.97, AVE = 0.94; Vendor: CA = 0.96; CR = 0.98, AVE = 0.96) ^b
Contextual ambidexterity ^{c, d}	(Customer: CA = 0.99, CR = 0.99, AVE = 0.94; Vendor: CA = 0.99; CR = 0.99, AVE = 0.90) <i>Alignment</i> 1. The management systems in this relationship work coherently to support the overall objectives of this relationship. (CAL1) 2. The management systems in this relationship facilitate the efficient utilization of resources. (CAL2) 3. The management systems in this relationship cause us to use resources on productive activities.* 4. The management systems in this relationship make people in this relationship work at cross-purposes because of conflicting objectives.* 5. The management systems in this relationship support achievement of efficiency-related objectives of this relationship. (CAL3) (Customer: CA = 0.93, CR = 0.96, AVE = 0.88; Vendor: CA = 0.96; CR = 0.97, AVE = 0.93) <i>Adaptability</i> 1. The management systems in this relationship encourage people to challenge outmoded traditions/practices/sacred cows.* 2. The management systems in this relationship are flexible enough to allow us to respond quickly to changes in our markets. (CAD1) 3. The management systems in this relationship evolve rapidly in response to shifts in our business priorities. (CAD2) 4. The management systems in this relationship facilitate reconfiguration of activities to respond to changes in the external environment. (CAD3) (Customer: CA = 0.95, CR = 0.97, AVE = 0.91; Vendor: CA = 0.90; CR = 0.94, AVE = 0.84)
Decision interdependence	1. The decisions made in the relationship frequently involve interactions with the partner. (DI1) 2. Our decisions in this relationship are relatively unaffected by decisions of the partner.* 3. Our decisions in this relationship are dependent on information shared by the partner. (DI2) 4. We collaborate with the partner for decisions pertaining to the relationship. (DI3) (Customer: CA = 0.87, CR = 0.92, AVE = 0.79; Vendor: CA = 0.72; CR = 0.83, AVE = 0.62)

Appendix B. Continued

Constructs	Measurement items
Use of operations support systems	For the everyday operations in the relationship for this supply chain service, our company relied on information systems support: 1. ...to conduct operational routines. (OSS1) 2. ...to execute a sequence of activities. (OSS2) 3. ...to comply with standardized procedures. (OSS3) (Customer: CA = 0.96, CR = 0.98, AVE = 0.93; Vendor: CA = 0.97; CR = 0.98, AVE = 0.94)
Use of interpretation support systems	For the managerial processes and decision-making purposes in this relationship, our company relied on IS support: 1. ...to generate inferences from past events (e.g., exceptions about supply chain services, and what worked and what did not). (ISS1) 2. ...to create new interpretations by combining information (e.g., uncovering trends and patterns about supply chain services). (ISS2) 3. ...to construct multiple interpretations (e.g., electronic dialogue to exchange different perspectives). (ISS3) 2. ...to represent information in multiple ways (e.g., slicing and dicing information based on requirements of different roles). (ISS4) (Customer: CA = 0.95, CR = 0.96, AVE = 0.86; Vendor: CA = 0.88; CR = 0.91, AVE = 0.73)

Notes. 1. All items were measured on a seven-point scale anchored by 1 = “strongly disagree” and 7 = “strongly agree.” 2. CA: Cronbach’s Alpha, CR: Composite Reliability, AVE: Average Variance Extracted. 3. We described for respondents that management systems for logistics IOR include service-level agreements, incentives, planning and review meetings, and methods for contract renegotiation. 4. Contextual Ambidexterity was calculated as the mean of nine items—i.e., the mean of the cross-products of the three alignment (CAL) items with the three adaptability (CAD) items.

*Items dropped in measurement refinement process due to low loadings.

Appendix C. Item Loadings from Principal Component Analysis

Construct	Items	Customer sample				Vendor sample			
		1	2	3	4	1	2	3	4
Contextual ambidexterity	CAL1 × CAD1	0.92	0.19	0.13	0.18	0.91	0.10	0.26	0.07
	CAL2 × CAD2	0.93	0.20	0.13	0.16	0.93	0.02	0.16	0.10
	CAL3 × CAD3	0.92	0.15	0.20	0.15	0.90	0.19	0.16	0.20
	CAL2 × CAD1	0.93	0.18	0.10	0.16	0.92	0.06	0.21	0.10
	CAL3 × CAD2	0.92	0.20	0.17	0.14	0.92	0.08	0.20	0.13
	CAL1 × CAD3	0.91	0.17	0.20	0.19	0.91	0.17	0.16	0.13
	CAL1 × CAD2	0.92	0.21	0.16	0.17	0.91	0.06	0.21	0.08
	CAL2 × CAD3	0.93	0.16	0.16	0.17	0.93	0.13	0.12	0.15
Decision interdependence	CAL3 × CAD1	0.93	0.17	0.14	0.15	0.90	0.11	0.25	0.13
	DI1	0.26	0.14	0.16	0.82	0.24	0.08	0.25	0.81
	DI2	0.12	0.14	0.14	0.88	0.38	0.31	−0.12	0.52
Use of interpretation support systems	DI3	0.32	0.18	0.16	0.79	0.05	0.07	−0.18	0.89
	ISS1	0.17	0.79	0.38	0.16	0.11	0.43	0.73	0.07
	ISS2	0.22	0.90	0.21	0.14	0.40	0.34	0.60	−0.06
	ISS3	0.23	0.88	0.19	0.16	0.52	0.11	0.71	−0.14
Use of operations support systems	ISS4	0.27	0.86	0.21	0.12	0.30	0.21	0.86	0.02
	OSS1	0.20	0.27	0.88	0.17	0.04	0.93	0.22	0.08
	OSS2	0.21	0.33	0.87	0.19	0.10	0.94	0.20	0.14
	OSS3	0.28	0.29	0.86	0.17	0.19	0.90	0.27	0.10

Notes. 1. The items are listed in Appendix B (Measurement Items for Constructs). 2. The item loadings were obtained using principal component analysis with varimax rotation. 3. The sample sizes for the customer and vendor data are 238 and 76, respectively.

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